

Jihoi Kim

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EDUCATION

PhD in Mathematics

Advisor: Pierre Raphaël

University of Cambridge

MA in Mathematics

Pass with Distinction

University of Cambridge

BA in Mathematics

First Class Honours

University of Cambridge

RESEARCH INTERESTS

Singularity formation and long-time dynamics in nonlinear PDEs, with a focus on the construction and stability of self-similar and blow-up profiles across dispersive and fluid models. Core techniques include spectral analysis of linearized operators, phase-portrait methods for profile construction, and vacuum-adapted energy methods; model problems include the semilinear wave equation and the compressible Euler equations with physical vacuum.

PUBLICATIONS AND PREPRINTS

1. On self-similar blow-up for the energy supercritical semilinear wave equation

Journal de l'École polytechnique — Mathématiques

DOI: 10.5802/jep.282

2. On smooth self-similar solutions to the compressible Euler equations with physical vacuum

Preprint

arXiv:2608.30443

3. On the stability of blow up profiles in compressible Euler equation with physical vacuum

With Mahir Hadzic (UCL)

Work in progress

4. On the C^∞ integral curve of a nonlinear ODE

With Pierre Raphaël

Preprint in preparation (not yet on arXiv)

TALKS

On self-similar blow-up of the energy supercritical wave equation

2024

Online Seminars on PDEs and Harmonic Analysis, KAIST

On the regularity of self-similar profiles

2025

Nonlinear Waves and Hamiltonian PDEs, Courmayeur

TEACHING EXPERIENCE

Example Class, Analysis of PDEs

University of Cambridge

Mathematical Tripos Part III

Supervisions

Analysis and Topology; Asymptotic Methods